Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 17	
TITLE: Swing and Applets	

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

1. **Create a simple Student Registration Form GUI using Swing.**

Programme

StudentRegistration.java 1

C: > Users > RIKISHA > PROGRAMMING > Java > src > StudentRegistration.java > StudentRegistration

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class StudentRegistration extends JFrame {

    JLabel idLabel, nameLabel, courseLabel, ageLabel;
    JTextField idField, nameField, courseField, ageField;
    JButton submitButton;

    StudentRegistration() {

        setTitle(title: "Student Registration Form");
        setSize(width: 400, height: 300);
        setLayout(new GridLayout(rows: 5, cols: 2, hgap: 10, vgap: 10));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Labels
        idLabel = new JLabel(text: "Student ID:");
        nameLabel = new JLabel(text: "Name:");
        courseLabel = new JLabel(text: "Course:");
        ageLabel = new JLabel(text: "Age:");

        // Text fields
        idField = new JTextField();
        nameField = new JTextField();
        courseField = new JTextField();
        ageField = new JTextField();

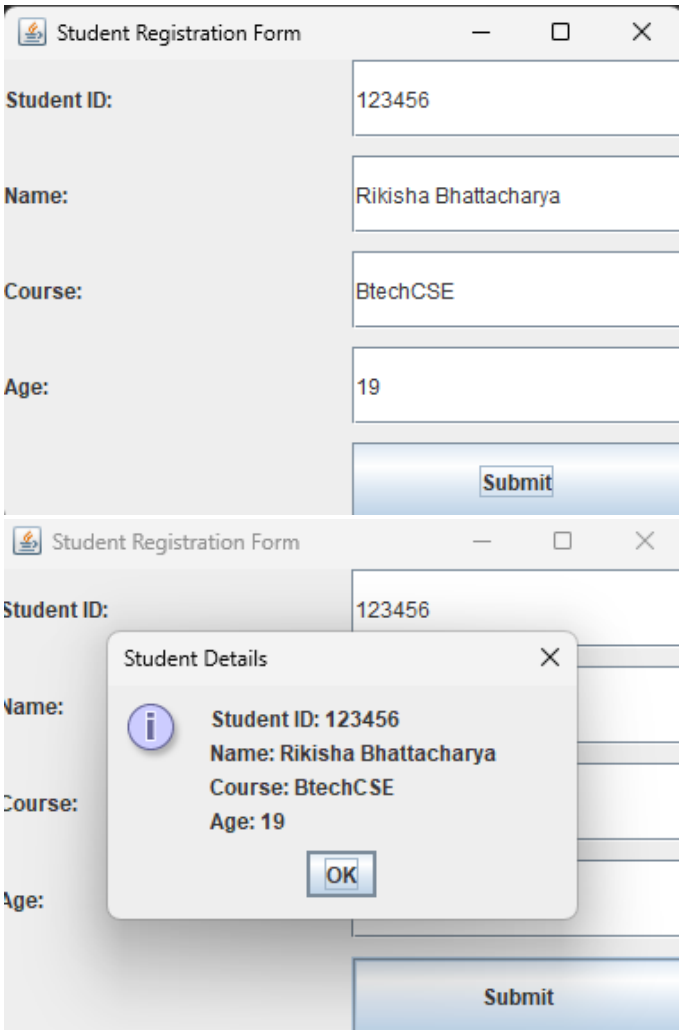
        // Button
        submitButton = new JButton(text: "Submit");

        // Add components
        add(idLabel);
        add(idField);

        add(nameLabel);
        add(nameField);
```

```
39
40          add(courseLabel);
41          add(courseField);
42
43          add(ageLabel);
44          add(ageField);
45
46          add(new JLabel(text: ""));
47          add(submitButton);
48
49          // Button action
50          submitButton.addActionListener(new ActionListener() {
51              public void actionPerformed(ActionEvent e) {
52
53                  String id = idField.getText();
54                  String name = nameField.getText();
55                  String course = courseField.getText();
56                  String age = ageField.getText();
57
58                  JOptionPane.showMessageDialog(
59                      StudentRegistration.this,
60                      "Student ID: " + id +
61                      "\nName: " + name +
62                      "\nCourse: " + course +
63                      "\nAge: " + age,
64                      title: "Student Details",
65                      JOptionPane.INFORMATION_MESSAGE
66                  );
67              }
68          });
69
70          setVisible(b: true);
71      }
72
      Run | Debug
73      public static void main(String[] args) {
74          new StudentRegistration();
75      }
76  }
```

Output



2. Create a simple Employee Registration Form using Java Swing GUI that takes employee details such as Employee ID, Name, Department, and Salary. Display the entered information in a dialog box.

Programme

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class EmployeeRegistration extends JFrame {

    JLabel idLabel, nameLabel, deptLabel, salaryLabel;
    JTextField idField, nameField, deptField, salaryField;
    JButton submitButton;

    EmployeeRegistration() {

        setTitle(title: "Employee Registration Form");
        setSize(width: 400, height: 300);
        setLayout(new GridLayout(rows: 5, cols: 2, hgap: 10, vgap: 10));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Labels
        idLabel = new JLabel(text: "Employee ID:");
        nameLabel = new JLabel(text: "Name:");
        deptLabel = new JLabel(text: "Department:");
        salaryLabel = new JLabel(text: "Salary:");

        // Text fields
        idField = new JTextField();
        nameField = new JTextField();
        deptField = new JTextField();
        salaryField = new JTextField();

        // Button
        submitButton = new JButton(text: "Submit");

        // Add components
        add(idLabel);
        add(idField);

        add(nameLabel);
        add(nameField);

        add(deptLabel);
        add(deptField);

        add(salaryLabel);
        add(salaryField);

        add(new JLabel(text: ""));
        add(submitButton);

        // Button action
        submitButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {

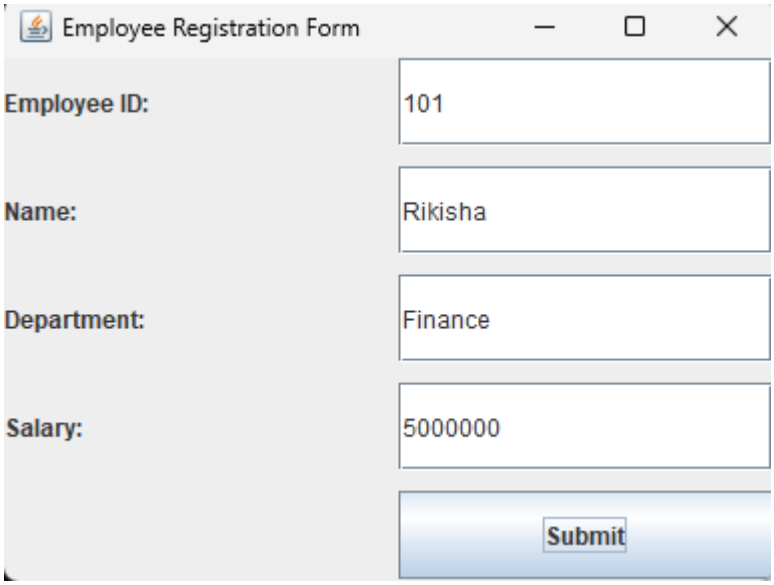
                String id = idField.getText();
                String name = nameField.getText();
                String department = deptField.getText();
                String salary = salaryField.getText();

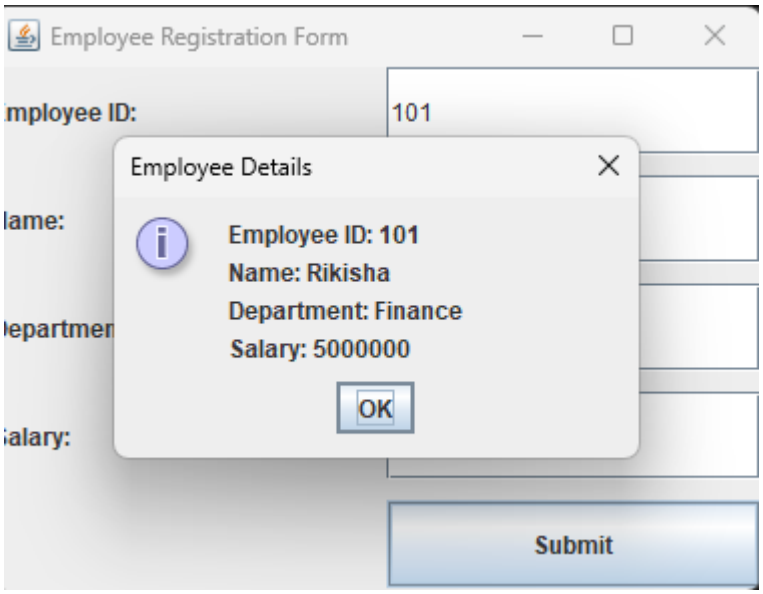
                JOptionPane.showMessageDialog(
                    EmployeeRegistration.this,
                    "Employee ID: " + id +
                    "\nName: " + name +
                    "\nDepartment: " + department +
                    "\nSalary: " + salary,
                    title: "Employee Details",
                    JOptionPane.INFORMATION_MESSAGE
                );
            }
        });

        setVisible(b: true);
    }

    Run | Debug
    public static void main(String[] args) {
        new EmployeeRegistration();
    }
}
```

Output





Conclusion:

Thus, the use of Java Swing components and Applets was successfully demonstrated by creating Student and Employee Registration Form GUIs. The programs helped in understanding how graphical user interfaces can be designed using components such as labels, text fields, buttons, and dialog boxes. The assignment provided practical knowledge of developing interactive and user-friendly GUI applications in Java.